

Diligently Create The Cradle Of Life

Baby Incubator

AI-1B

(V3.5)

User Manual

**Please read and keep this instruction in
order to use the Infant radiant warmer correctly**

READ AND KEEP THIS MANUAL CAREFULLY PLEASE

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1 Summary

It is the advanced technology in such disciplines as comprehensive clinical medicine, machinery, computer automatic control, sensor, etc. that the baby in incubator, offers newborn baby and disease baby to a fine environment where air purify, temperature and humidity is suitable, similar to uterus of the parent. The temperature of cabin space of the baby in incubator should be set up on the doctor's order, Baby's skin temperature, baby's air temperature of the cabin and baby's humidity of the cabin have digital display, when baby's skin temperature and air temperature of the cabin over the normal value or present other unusual situations, there is alarming (power cut off, fan failure, sensor trouble, air temperature over, skin temperature over, deviation between display temperature and setting up temperature, etc.), guarantee the dependability and security of the one machine.

This manual includes two parts of operation instruction and technical specification, and is suitable for AI-1B Baby incubator to install, use, maintain common trouble analysis and get rid of, etc.

1.1 Definitions and Symbol

1.1.1 Definition

1、 Air temperature control

The air temperature of baby's compartment is automatically controlled by an air temperature sensor close to a value set by the user.

2、 Incubator temperature

The temperature of the air at a point 10 cm above the center of the mattress surface in the baby compartment.

3、 Control temperature

Temperature selected at the temperature control (the temperature needed of baby's compartment).

1.1.2 The symbol

	On of power switch		Attention, consult accompanying documents
	Off of power switch		To set the temp greater than 37°C
	Protective earth (ground)		Direct Current
	Fuse		Clockwise down, anticlockwise up
	Turn off(function switch)		Alternating current
	Bilateral rotation		

	To set the temp greater than 37°C		Audio paused
	Temperature setting increase button		Temperature setting reduce button
	Keyboard lock key		Alarm reset button
	Alarm reset button		Low priority alarm
	Battery		ON (function switch)
	Humidification regulator		Humidification power low gear
	Humidification power medium gear		Humidification power high gear

1.2 Scope of application

This product is used to provide a suitable environment for premature infants and sick infants.

1.3 Structure composition and characteristics

1.3.1 Structure and composition

AI-1B is composed of hood, control cabinet and cabinet.

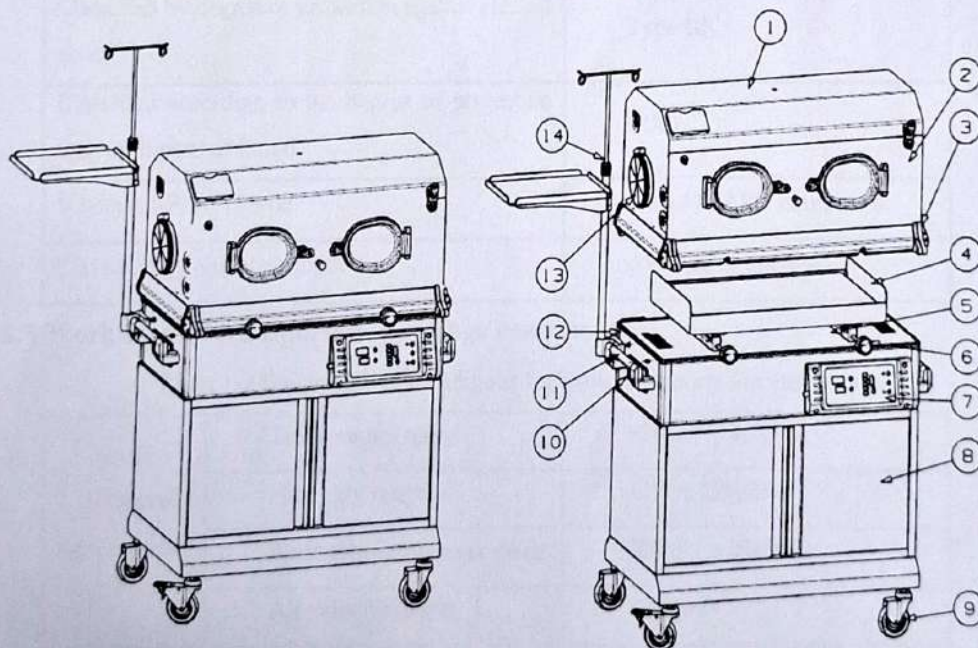


Fig 1-1 Incubator structure diagram

- | | | |
|-----------------|---|--------------------|
| 1.Thermostat | 2.Front gate | 3.Front door hinge |
| 4.Crib | 5.Pallet | 6.Shaker |
| 7.Control box | 8.Cabinet | 9.Casters |
| 10.Water tank | 11.Handle | 12.Control cabinet |
| 13.Swing window | 14.Infusion support and tray components | |

1.3.2 Features

- 1、The product has a power-on self-check function, and a variety of fault alarms.
- 2、Air temperature control mode, separate screen display of Air temperature/heating power percentage.
- 3、Five kinds of audible and visual alarm functions.
- 4、 $>37^{\circ}\text{C}$ temperature setting function.
- 5、Humidity is adjustable continuously
- 6、The inclination of infant bed is adjustable.
- 7、Plug-in axial flow DC fan
- 8、Magnesium-aluminum alloy control cabinet, fluorine spraying process to prevent alkali and acid, paint and rust.

1.4 Equipment classification

Chart 1-1 Equipment Type

Classified by type of electric shock prevention	Class I
Classified by degree of protection against electric shock	Type BF
Classified according to the degree of protection against ingress of liquid	IPX0
Whether AP, APG type	Non-AP, APG equipment
Classified by operating mode	continue to operate

1.5 Working and transportation storage conditions

Chart 1-2 Environmental conditions for work, transportation and storage

Operating conditions	Temperature range	$+20^{\circ}\text{C} \sim +30^{\circ}\text{C}$
	Humidity range	$30\% \sim 75\%\text{RH}$
	Atmospheric pressure range	$700\text{hPa} \sim 1060\text{hPa}$
	Air velocity of rate	$<0.3\text{m/s}$
transport	Temperature range	$-10^{\circ}\text{C} \sim +55^{\circ}\text{C}$

	Humidity range	$\leq 95\%$
	Atmospheric pressure range	500hPa~1060hPa



The outer packaging box of the product can be transported by general transportation means, and it should be handled with care when transporting, and it should be kept upright and moisture-proof during transportation.

1.6 Period of use

- 1、Product life: 8 years.
- 2、Production date: See product label.

2 Working principle

2.1 The principle of constant temperature and air convection circulation system

Baby Incubator body is divided into two stories. There are two rectangular holes on the baby compartment, then it will form air convection circulatory system from head to foot. The controller is equipped with the axle flowing blower, heated tube and the water trough; the baby compartment is equipped with the hood, it is convenient to take down the crib. Under the driving of the axle flowing blower, the heat and moisture circulate along such a direction from top to bottom, from left to right in the baby incubator, and then reach the balance of the temperature and the humidity. In the space of the axle flowing blower where the air pressure is negative, there is a hole having a diameter of 13 mm, from where a small amount of air outside enter the baby incubator after purifying, achieve the purpose that CO₂ does not exceed standard.

2.2 Principle of temperature control and display

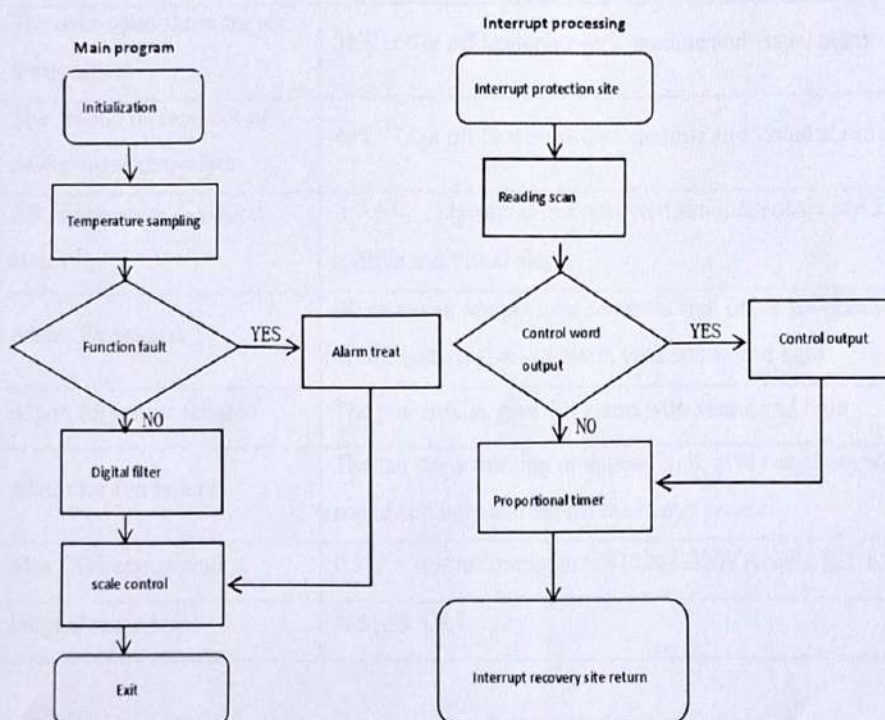


Fig 2-1 Procedure block diagram

3 Performance

Chart 3-1 Main performance index

Model	AI-1B
Power supply	AC 110V $\pm$ 11V; 50Hz $\pm$ 1Hz
Power consumption	650VA (including power output socket: 200VA)
Temperature display range	Air temperature: 10 $^{\circ}$ C $\sim$ 42 $^{\circ}$ C
Temperature control range	Air temperature: 25 $^{\circ}$ C $\sim$ 37 $^{\circ}$ C; 37.1 $^{\circ}$ C $\sim$ 38 $^{\circ}$ C
Temperature fluctuation	$\pm$ 0.5 $^{\circ}$ C
Temperature display accuracy (resolution)	0.1 $^{\circ}$ C
Temperature uniformity	$\leq$ 0.8 $^{\circ}$ C
Warm-up time	$\leq$ 45min
The over-temp alarm for air temperature	38 $^{\circ}$ C (Cut off heater power) audible and visual alarm
The second thermal cut-off device over-temp alarm	40 $^{\circ}$ C (Cut off heater power) audible and visual alarm
Air temperature deviation alarm	$\pm$ 3.0 $^{\circ}$ C (Heater is disconnected simultaneously at +3.0 $^{\circ}$ C) audible and visual alarm
Alarm for sensors	When the air temperature sensor is shut off or shot circuit, cut off the heater, give out alarm with sound and light
Alarm for power failure	The power fails, give out alarm with sound and light
Alarm for fan failure	The fan stops working or appear fault, give out alarm with sound and light and cut off the heater power
Max CO2 concentration	0.3% (Test according to GB11243-2008 Article 105.1)
Internal noise level	$\leq$ 55dB (A)

4 Function description

4.1 Operating mode

The incubator has a temperature control system in the air temperature mode : the output of the heater is automatically controlled according to the temperature detected by the air temperature sensor in the cabinet.

4.2 The main function

1、The handle of incubator: When need move baby incubator, pull or push it by the handle at the two side of incubator.

2、Front door lock : Press the left knob handle until it is close to the front door, and then turn the handle to the left to make the pressure bar vertical; press the right knob handle to close the front door, and then rotate the handle to the right to make the pressure bar vertical, and then press the self-locking door lock To open the front door. Close the front door to lock the self-locking door lock, and then press the left and right handles until they are close to the front door, turn the left knob to the right, and the right knob to the left to the level of the pressure lever to achieve the safety lock of the front door.

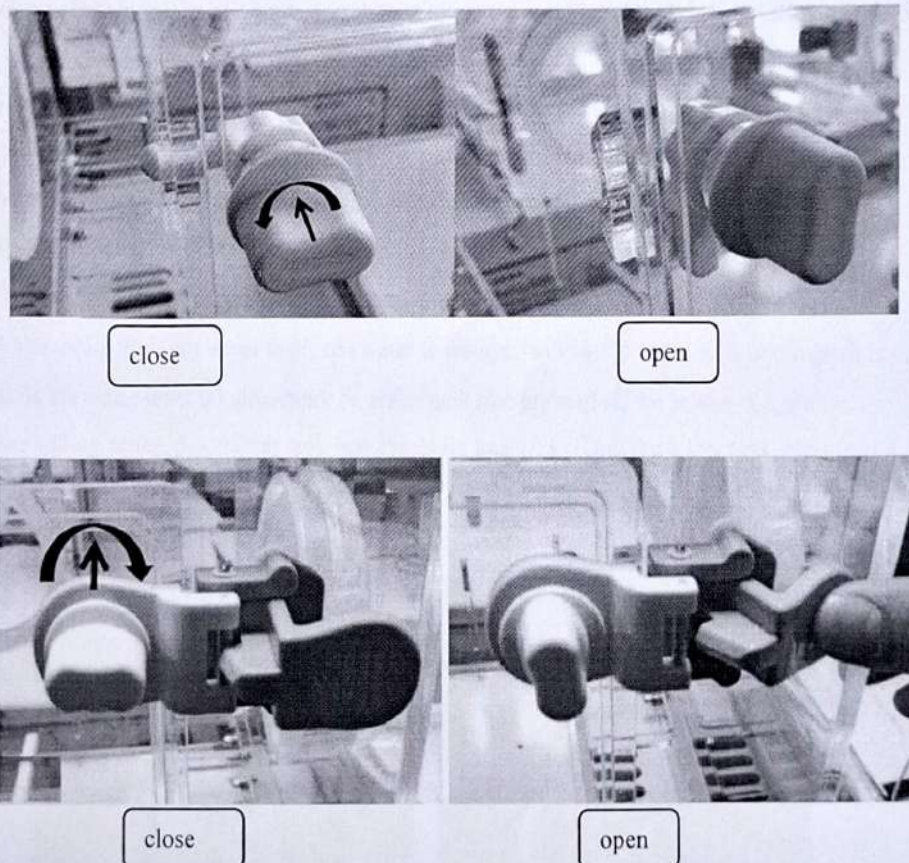


Fig 4-1 Open the front door operation

3. Oval operation windows: There are operating windows at hood's front and back door, so as two (or more than two) medical workers to go on with the baby through operation window. While opening the window, the medical worker only needs to press the lock organization with the finger, the window goalkeeper opened automatically.

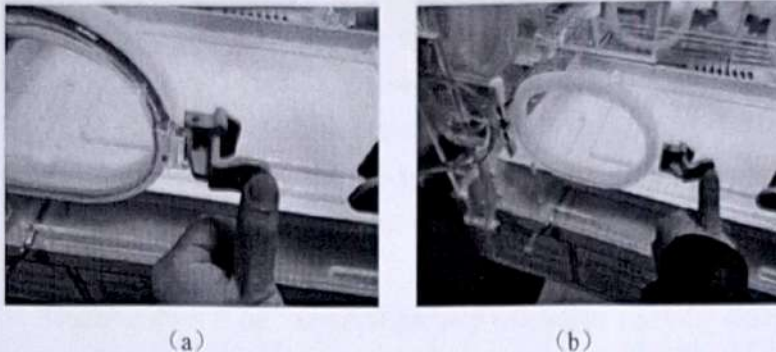


Fig 4-2 Open the oval operation windows operation diagram

Iris ports (circle operation window): There are circle operating window on the side for infusion or ventilation pipeline use. When using, first put one end of the special cloth cover on the inner side of the thermostatic cover, and cover the other end on the outer side (to penetrate the infusion or ventilation pipe, or not to wear), then rotate the outer handle of cloth cover, make cloth cover ventage reduce to closed.

Support locking plate: When the hood is lifted upwards, the side support locking plate will limit the raised height and play a supporting role; when closing the hood, the support locking plate must be pressed down, and the hood can be closed after the self-locking screw is out of the limit.

Sealing rubber ring: Medical worker's hands feel comfortable while stretching into and operating in the window, play a sealed role after the window is shut off.

Water tank: Pull out water tank, add water to trough. How much water adds to trough, it is depend on the water level window mark on water tank (the highest: H, the lowest: L). after finishing adding water, push water tank into incubator totally. Pull out the water tank, can outwell water cleanout it.

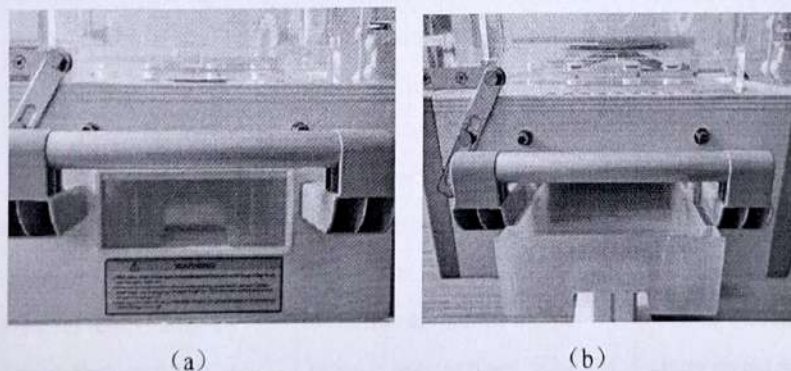


Fig 4-3 Open the water tank operation

8、I.V. Pole: While needing for baby's infusion, used for hanging the infusion jar, the largest loading capacity is 20N

4.3 Alarm

4.3.1 Summary

In order to ensure that the incubator provides a suitable temperature and humidity environment for the newborn baby, in the use and maintenance of the equipment, the user can be notified in time when there is a failure, so that it can quickly take effective measures to reduce and avoid harm to the patient. The equipment is specially designed with the following 5 kinds of alarm status (see Chart 4-1), which are divided into physiological alarm status and technical alarm status:

1、Technical alarm status: Sensor alarm, over-temperature alarm, fan alarm and air temperature deviation alarm;

2、Other alarm status: Power outage alarm.

Chart 4-1 Alarm status description

Alarm status	Alarm activation conditions	Status description	Delay description
Power failure alarm	When the power switch of the device is turned on, if the power supply to the device is interrupted, the device will sound an audible and visual alarm for at least 10 minutes, or until the power supply is restored.	When the device emits a "beeps." audible alarm, and the "power off" red indicator light on the control panel flashes, all other digital displays and indicators are off. The device enters the "power interruption" alarm state.	The average alarm state delay for power failure is 1 second, and the average alarm signal generation delay is 1 second. The operator is located within 1 meter in front of the equipment.
Air temperature deviation alarm	Under the control of the air temperature (required to set the temperature to be at least 3° C higher than the ambient temperature), after the temperature is stable, if the test temperature differs from the set temperature by more than $\pm 3^{\circ}\text{C}$, the incubator will	When the device emits a "beeps" audible alarm, and the control panel will not light up except for the yellow indicator light for "deviation". The device enters the temperature "deviation" alarm state. Alarm in air temperature control mode, that is, air temperature deviation alarm, the alarm range is greater than $\pm 3^{\circ}\text{C}$. If the difference between the	The maximum alarm status of the air temperature deviation is delayed by 1.5 seconds, and the maximum alarm signal generation delay is 1 second. The average alarm status delay of the temperature deviation of the cabinet is 1 second, and the average alarm signal generation delay is 1 second.

	automatically send out an audible and visual alarm to remind The operator takes measures. If the difference is more than +3℃, the equipment will automatically cut off the heater power supply.	corresponding display temperature and the set temperature is positive, it is a positive deviation, otherwise it is a negative deviation.	The operator is located within 1 meter in front of the equipment.
Sensor failure alarm	If the air temperature sensor is open or short-circuited, the device will give an audible and visual alarm and cut off the heater's power supply at the same time.	When the device emits "beeps" an audible alarm, except for the "sensor" red indicator on the control panel, the other alarm indicators are off , The air temperature display window displays "Err", indicating that it has entered the "air temperature sensor failure" state;	The maximum alarm status of the air temperature is delayed by 3 seconds, and the maximum alarm signal generation delay is 1 second. The average alarm status delay of the air temperature is 2 seconds, and the average alarm signal generation delay is 1 second. The operator is located within 1 meter in front of the equipment.
Fan failure alarm	In any case, if the fan stops or the fan sensor system fails, the device will emits an audible and visual alarm and cut off the heater's power supply.	When the device emits "beeps" an audible alarm, except for the "fan" red indicator on the control panel, the other alarm indicators are off. , The temperature display window and the setting window digital tube are not bright. The equipment enters the "fan" alarm state.	The maximum alarm state of the fan is delayed by 16 seconds, and the maximum alarm signal generation delay is 1 second. The average alarm status delay of the fan is 12 seconds, and the average alarm signal generation delay is 1 second. The operator is located within 1 meter in front of the equipment.

Over temperature alarm	1) The device works in the air temperature control mode. When the air temperature is set below 37° C, after the temperature stabilizes, if the test temperature reaches 38° C, the device will automatically emit an audible and visual alarm and cut off the heater power.	When the device emits "beeps" an audible alarm, and the control panel except for the "over temperature" red indicator flashes and lights up, other alarm indicators No light, the air temperature display window displays the high actual temperature value. The device enters the "over temperature" alarm state.	The maximum alarm state of the air temperature over-temperature is delayed by 3 seconds, and the maximum alarm signal generation delay is 1 second. The average alarm state delay of over-temperature of the air is 2 seconds, and the average alarm signal generation delay is 1 second. The operator is located within 1 meter in front of the equipment.
	2) The equipment works in the air temperature control mode. When the air temperature is set at 37°C~38°C, after the temperature is stable, if the test temperature exceeds 40°C, the equipment will automatically emit an audible and visual alarm and cut off the heater power supply.		

4.3.2 Time to verify whether the alarm system is normal

After the initial installation of the incubator or the removal and reinstallation of the components due to cleaning and maintenance, and before use, it is necessary to verify whether the alarm system of the equipment is normal. See 6.5 for the verification method.

4.3.3 The priority of alarm status

According to the "Testing and Guidelines for Alarm Systems in Medical Electrical Equipment and Medical Electrical Systems" standards and incubator safety standards, the alarm status of the equipment must not only produce a visual alarm signal that meets the requirements, but also an audible alarm signal that meets the requirements. Ensure the safety of patients.

1、High priority alarm: Power failure alarm、fan alarm、Sensor alarm、Over-temperature alarm.

2、Low priority alarm : air temperature deviation alarm.

Note:

- a) The sound alarm indication of power interruption alarm is different from other high-priority alarms, and its source is a single buzzer.
- b) When the power loss time does not exceed 30s, the alarm settings before the power loss are automatically restored, and the power loss will not cause the alarm settings to be lost.
- c) The high priority alarm sound pressure level is greater than or equal to the low priority alarm sound pressure level, both of which are greater than 65dB.
- d) If there are multiple alarm states at the same time, the device will give an audible alarm to the alarm state with a higher priority and a visual alarm at the same time.
- e) Except the power interruption alarm, other alarms can be muted by the sound pause button, and the sound alarm will be automatically restored within 10 minutes. During the mute period, if you press the sound pause button again or long press the alarm reset button, the sound alarm can be restored.
- f) The over-temperature alarm is a latching alarm signal, and the over-temperature alarm state can be removed after the fault is removed and the machine must be restarted.

Warning: If the incubator is used in any independent area, there will be potential dangers if different alarm presets are used.

5 Installation

(Notice: For ensure safety, should at least 2 people finished the installation.)

※ Must use a wrench to remove the 4 fixing bolts at the bottom of the wooden box before removing the control cabinet components

5.1 Incubator installation

1、Install the 4 casters as shown in Figure 5-1 and tighten them with a 22mm open-end wrench.

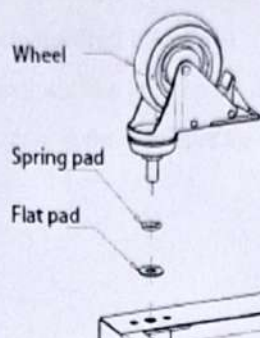


Figure 5-1 Caster installation diagram

2、First remove the locking screw with a flat-head screwdriver, then place the thermostatic cover on the control cabinet, and then pass the locking screw through the right-hand hinge of the thermostatic cover and lock it on the support and fixing block (as shown in Figure 5-2).



Figure 5-2 Diagram of fixed thermostatic hood

3. Fix the infusion rod to the infusion support of the control cabinet. (Fix firmly with screws)

4. Loosen the locking knob on the infusion pole, adjust the height and direction of the hook, and then tighten the locking knob; use the hexagonal wrench to loosen the 4 screws for the equipment equipped with the tray, and install the tray on the infusion pole to fix it tight.

5.2 Installation of air inlet and air outlet guardrail

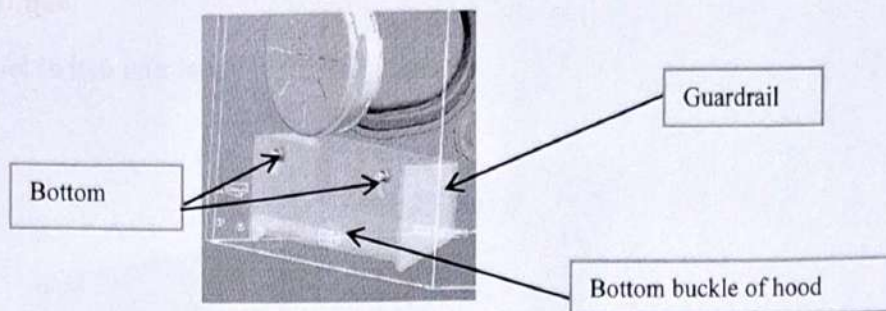


Figure 5-4 Guardrail installation

Note: The bottom of the air inlet guardrail is inclined

Match the round hole in the guardrail with the hook on the thermostat cover, and let the back slide. Slide the bottom of the rear guardrail to the bottom of the thermostat cap.

6 Operational use

6.1 Control panel switch and interface distribution

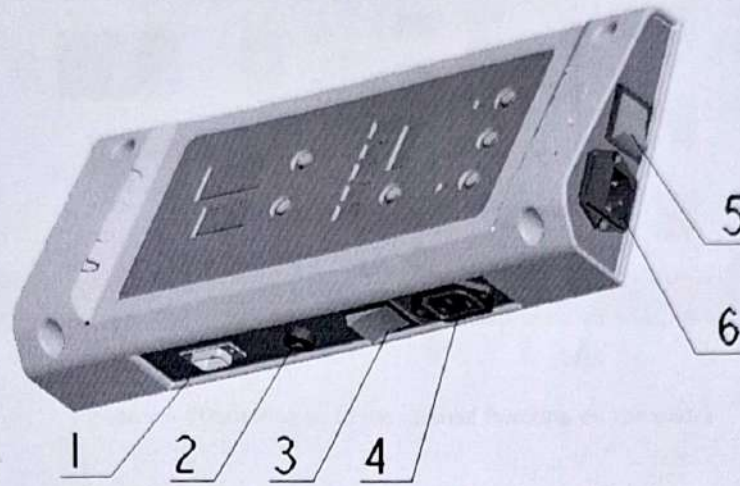


Figure 6-1 Control panel switch and interface distribution

- | | | |
|------------------|----------------------|-----------------------|
| 1. RS-232 port | 2. 1A Fuse holder | 3. Power switch |
| 4. Output socket | 5. Controller switch | 6. Input power socket |

Chart 6-1 Control function on the panel

Power switch	Power switch under the control panel, turning on the power switch is a prerequisite for the operation of this incubator.
Controller switch	The controller switch is on the right side of the control panel. Turn on the power switch and then turn on the controller switch to allow the incubator to work normally.
Auxiliary output power socket	The auxiliary output socket of the incubator power supply provides AC 110V/50Hz power supply, which can provide power for auxiliary equipment including phototherapy equipment. The maximum output power is 200VA.

6.2 Panel control function

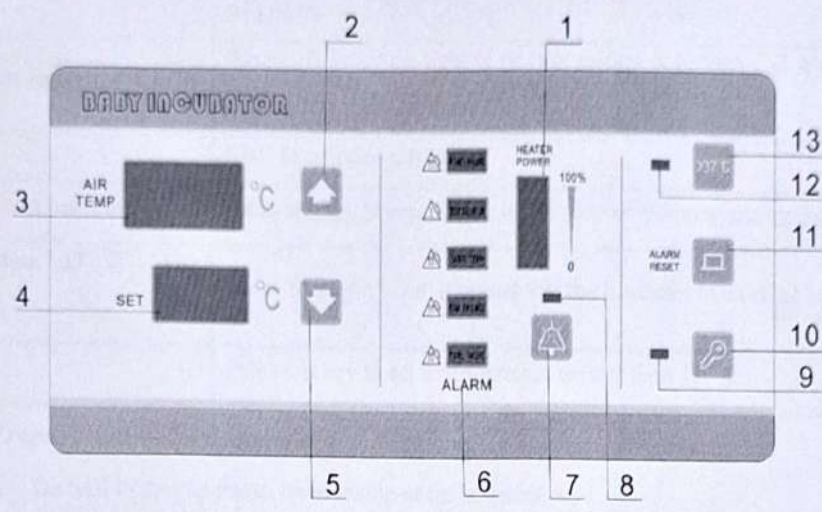


Figure 6-2 Distribution of the control function on the panel

- | | | |
|------------------------------------|--|--|
| 1. Heating power indicator | 2. Air temperature setting increase button | 3. Air temperature real-time temperature display |
| 4. Air temperature setting display | 5. Air temperature setting reduction key | 6. Fault alarm classification indicator |
| 7. Sound pause button | 8. Sound pause button indicator | 9. Key button indicator |
| 10. Key | 11. Alarm reset button | 12. More than 37°C key indicator |
| 13. >37°C | | |

Chart 6-2 Button description

1. Heating power indicator	Indicates the percentage of heating power.
2. Air temperature setting increase button	In the setting state, press this key to increase the setting value, and keep pressing this key to speed up the increase speed of the setting value.
3. Air temperature real-time temperature display	Display the temperature at the location of the air temperature sensor.
4. Air temperature setting display	Display the air temperature mode control value set by the operator.
5. Air temperature setting reduction key	In the setting state, press this key to decrease the setting value. Keep pressing this key to speed up the decrease of the setting value.
6. Fault alarm classification indicator	When an alarm occurs, the corresponding alarm type indicator light is on.
7. Sound pause button	When an alarm occurs, press this button to temporarily stop the audible alarm (except for the power failure alarm), and the audible alarm will automatically recover within 10 minutes.

8.Sound pause button indicator	When an alarm occurs, press the sound pause button and the indicator light will be lit.
9.Key button indicator	When the indicator light is on, the setting is locked, and when the light is off, the setting is activated.
10.Key	Lock or activate settings.
11.Alarm reset button:	Press this key to reset to the initial state or memory setting state.
12.More than 37 °C key indicator	When this light is on, it means that the incubator is working at >37° C.
13.>37°C	Press this key to set a temperature greater than 37° C.

6.3 Preparation before operation

- 1、Do well in the cleanness, disinfecting of the instrument.
- 2、The casters of the incubator are locked to prevent movement during work
- 3、Insert connector of the power cord in the equipment's power input socket, the other plug in AC 110V,50Hz socket (earth wire must be firm and reliable).

6.4 The operation method

1、Start the machine

Turn on the switch, then turn control switch on, the controller enters self-examine state, and then the instrument enters normal state:

when "key" button indicator is bright, the instrument is in air temp control mode, "set" window show temp set value, equipment take auto heating control as the temp of set, air temperature window real show incubator temp value.

2、Set control temp

According to the need of the clinical, when it is needed to change temperature value, please do as follows:

- a) Press the "key" button; remove the locking of the "set" button
- b) Press the "increase" button or the "reduce" button to set the temperature value; at foundation of 37°C, press the ">37°C" button (after the indicator is on), then press the "increase" button or the "reduce" button to set the control temperature value greater than 37°C.

3、Deal with the audible and visual alarm

When appearing the unusual situation or the instrument breaks down, the controller will classify and give out sound and light alarm automatically, cut off the heater power at the some time. Press "sound pause" button to make the alarm pause, and then check out the cause and remove it, the instrument will return to the normal state again (when alarming for over temp, must press "reset" button to let the instrument work again).

4. Adjusting for tilt of an Baby bed

Note: Changing the tilt and height of baby bed may influence temperature uniformity in baby compartment.

Shake Baby bed handle, can rising (or falling) the Baby bed or adjusting its tilt to meet clinical needs. After finished, make the handles stand in lower position that is easy to other operation.

5. The usage of humidity regulator

When using a humidity regulator, you must fill the water tank with distilled water and check the water level regularly. Since the humidity in the incubator is greatly affected by environmental conditions, the humidity knob should be adjusted during use to select the appropriate humidity.

First, turn the humidification switch clockwise, When the knob reaches the position , the humidification function is turned on, (Reference position of the switch: winter ; Summer ; Spring, Autumn ), After humidity stable (>2 hours), check the humidity of incubator, if not meet the requirement, can adjust rotary knob, clockwise can increase, otherwise reduce.

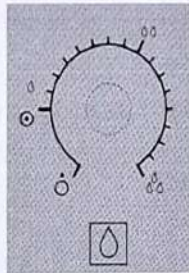


Figure 6-3 Humidification switch

6. System temperature parameter correction and its correction method (the user should not correct it if the test conditions are not available!)

Press and hold the "key" button for about 6 seconds, wait until the temperature setting display window displays "---", the first digit is flashing, press the "increase" button to make the first digit "1"; press the "decrease" button again, the second digit To flash, press the "increase" key twice to make the second digit "2"; press the "decrease" key to make the third digit flash, and press the "increase" key three times to make it "3". Then press the "key" button to enter the function main menu, and the display window displays "Pr.-".

Press the "increase" key to make the display window display Pr.1, and press the "key" key to confirm, it will enter the temperature error and over-temperature alarm correction submenu. The temperature display window displays $P \times \times$, and the adjustable display range is 0-4.0 (initial value is P2.0). In this state, you can press the air temperature setting "increase" button to modify the temperature display value of the air temperature and the standard thermometer display value Deviation (maximum is $\pm 2^{\circ}\text{C}$).

7、Oxygen operation (only applicable to models with oxygen delivery port)

When inputting oxygen, an oxygen monitor (not provided with this device) must be used to monitor the oxygen concentration!

Put one side of oxygen therapy connect oxygen flow meter (not offer), other side connect oxygen therapy of incubator. Adjust the incubator oxygen concentration by input oxygen flow (liter/minutes).

Chart 6-4 gives the reference value of oxygen concentration in the incubator.表 6-4 oxygen concentration reference value

Supply oxygen (L/M)	1	2	3	4	5	6	7	8
Approximately oxygen concentration (%)	28	30	34	39	42	46	50	53

Note: The oxygen concentration values in Chart 6-4 are measured in the infant compartment after being stabilized, and can only be used as a reference for oxygen delivery. The actual oxygen concentration can only be monitored regularly by a calibrated oxygen monitor located in the infant compartment. The optimal oxygen concentration value must be given by the attending physician according to the actual partial pressure of oxygen in the patient's blood.

The use of oxygen will increase the risk of fire. Auxiliary equipment that can generate sparks should not be placed in the incubator. Before configuring the equipment to use oxygen, turn off the switch of the phototherapy light box (cold light source).

6.5 Examination of the function

Every time when reinstallation incubator after disassembly parts for cleanout or maintain, should inspection incubator function, ensure incubator normal work

1、The inspection of alarming for power failure.

Before the equipment plug insert in the AC power, turn on control switch and power switch, the equipment will have audio and video alarm indicate "power failure", turn off these switches, the alarm will be removed.

2、Air temperature

Press the "key" button (the indicator light of the "key" key is off), the number in the air temperature setting window flashes, and then press the air temperature setting "increase" or "decrease" key to set the temperature.

3、The alarm of temp deviation

Set the air temp to 35°C, after temperature is constant, opened the equipment's front door, make the temperature of baby compartment drop, when it is lower than 32°C, the equipment will give out alarm with sound and light. Close the front door, after resuming in temperature, the alarm for temp deviation will be silent automatically.

4、 The alarm of over temperature

a) Press the "key" button of the incubator two seconds, the air temperature setting window shows "--" and the first character glitters. Press the "increase" make the first character is "1"; press "decrease" button ,make the second character flash ,press two times "increase " key, make second character is "2", press "reduce" key, the third character glitter, press three times "increase" key, make it as "3".then press "set" key ,into main menu, this display window show "Pr.-"

b) Press the "increase" (or "decrease") button to select "Pr.4", then press the "key" , this window show "OFF", press "increase"key, this window show "on" press "key" button exit menu 4, press again equipment will start again, after restart," over temp" indicator will flash, temperature-controlled instrument simulation loss control state. When the air temp rise to 38°C in temperature, will have audible and visual alarm, (at the same time switch off the heated power automatically).power off the equipment wait temp fall to normal temp restart it. Equipment auto exited over temp alarm test state.

5、 The alarm of the second cut-out temp.

Reference 4. a) into main menu, press the "increase" (or "decrease") button to select "Pr.5", then press the "key" , this window show "OFF", press "increase"key, this window show "on" press "key" button exit menu 5, press again equipment will start again, after restart," over temp" indicator will flash, temperature-controlled instrument simulation loss control state. When the air temp rise to 39.5°C in temperature, will have audible and visual alarm, (at the same time switch off the heated power automatically).power off the equipment wait temp fall to normal temp restart it. Equipment auto exited over temp alarm test state.

6、 The alarm of Fan failure

Try to prevent the fan from rotating. After about 5 seconds, the equipment will send out an audible and visual alarm and stop heating. The alarm will be automatically released after the fan resumes rotation.

7 Precautions

- 1、An incubator should be used only by appropriately trained personnel and under the direction of qualified medical personnel familiar with currently known risks and benefits of incubator use. That must read "manual" carefully before using the machine.
- 2、When starting the machine, must set the temperature value firstly, and can use after preheating for 60 minutes.
- 3、This machine has no water shortage alarm device, always observe the water level height of the tank, do not make the water shortage in the tank. It is recommended that users use distilled water for humidification and use a hygrometer to monitor the humidity in the incubator.
- 4、If the power fails, don't turn on the power switch for a long time, otherwise a power interruption alarm will occur and battery power will be consumed.
- 5、The incubator should not be placed in the position of the air outlet (such as the air conditioner). It is forbidden to block the air outlet and air inlet of the infant's room.
- 6、The incubator should not be placed in the position of the air outlet (such as the air conditioner). It is forbidden to block the air outlet and air inlet of the infant's room.
- 7、When using the incubator, lock the casters to prevent the equipment from moving.
- 8、The air temperature sensor needs to be calibrated every six months, and the calibration work must be carried out by authorized and qualified maintenance personnel.
- 9、When the front door of the hood is closed, the two front door handles should be rotated in place to make them jam in the positioning groove. When the front door is opened, the temperature displayed on the temperature display window may not be the actual temperature in the incubator, which may affect the temperature of the patient. Therefore, do not leave the front door open for a long time.
- 10、The power supply of the incubator uses F6.3AL 250V and F2AL 250V fuse tubes. When replacing, the power plug must be unplugged to ensure that the power supply is interrupted.
- 11、The air purification filter cloth should be checked and cleaned in time to avoid poor air intake and increase the concentration of carbon dioxide(CO₂) in the baby cabin.
- 12、The surface of the plexiglass hood and phototherapy lamp shall not be scrubbed with organic solvents such as alcohol, and the plexiglass hood shall not be exposed to direct ultraviolet radiation.
- 13、Baby bed tray load: The total weight of baby and items in the bed shall not exceed 100N.
- 14、When the equipment is abnormal, do not use it reluctantly, and use it only after professionals have found out the cause and solved the problem.
- 15、The control circuit of the incubator is connected with a relay, which will generate a small amount of electromagnetic radiation when the relay is activated. When using other auxiliary equipment, it is recommended to use good grounding and shielding measures.

16、 Before filling the water tank, turn off the humidity control switch or the power switch. After waiting for about 2 minutes, pull the water tank out for about 15cm continuously and steadily, fill the water tank with distilled water. After adding to the H water level, push the water tank slowly and continuously into place, so as not to damage the humidifying parts. Note: This equipment does not have the function of reducing humidity! Do not use when the ambient humidity is higher than 75%RH.

17、 The safety requirements of the auxiliary equipment of the equipment shall comply with the first part of GB 9706.1-2007 Medical Electrical Equipment: General Safety Requirements.

18、 The disposal of product waste must comply with the requirements of laws and regulations.

19、 Do not use it in an environment with portable and mobile radio frequency communication equipment, so as not to affect the normal operation of the product.

20、 This equipment service life is 8 years. When the service life expires, the product must be scrapped in time, and the disposal of product waste must comply with the requirements of laws and regulations.

8 Warning

1、 The incubator is not an AP/APG device, no oxygen delivery equipment and devices are attached. If users equip their own equipment for oxygen delivery, please pay special attention to:

a) An oxygen analyzer must be equipped during oxygen delivery and it is recommended to use an oxygen hood. The oxygen (volume) concentration of the air in the incubator should not exceed 25%. For operations, please refer to the oxygen analyzer instruction manual and similar documents.

b) When oxygen is given, it will increase the risk of fire. At this time, auxiliary equipment that generates sparks should not be placed in the incubator.

c) When the oxygen is switched on, even a small amount of flammable agents such as ether and alcohol left in the incubator can cause a fire.

d) The oxygen delivery operation will increase the noise level for the baby within the incubator.

2、 The AC power must be single-phase and three wires, the earth wire must be reliable.

3、 The incubator should be put in the clean, temp and humidity having a small change workplace.

4、 The incubator should be protected from direct sunlight and away from radiant heat sources to prevent the temperature of the incubator from rising to a dangerous level.

5、 The incubator is CLASS I and TYPE BF applied part equipment, particular care must be taken to ensure that additional equipment connected to the baby is electrically safe. That link with baby is set up must be grounded additionally it is reliable or insulating.

6、 The mode of operation of the equipment is CONTINUOUS OPERATION.

7、 Do not place or use materials that generate harmful gas or dust in the infant cabinet.

8、 When the auxiliary equipment uses the output socket, the rated power of the load shall not exceed the rated value of the output socket.

9、 When a certain function of the incubator is lost or malfunctions, the baby's treatment should be immediately terminated and shut down, and repairs should be performed by maintenance personnel approved and authorized by the company.

9 Clean and Maintenance

9.1 Cleaning

The incubator must be thoroughly cleaned, sterilized, and disinfected every time an infant is incubated (at least once a week for the same infant).

First, dip all incubator parts into sterilization solution, next rinse them with sterilized, clean and warm water. And then dry them completely. Finally wipe off moisture by the use of a soft cloth. Use disassembly methods of each part for cleaning and sterilization. Comply with the following methods for incubator cleaning and sterilization.

0.2~0.5% Benzalkonium chloride solution

0.2~0.5% Benzethonium chloride solution

0.02~0.05% Chlorhexidine solution

Note: If a solution is used for disinfection, it must be diluted before it can be used.

Undiluted solution cannot be used for cleaning and disinfection!

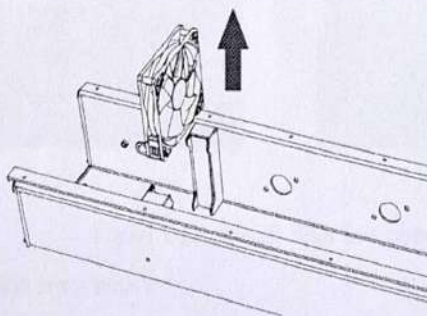
1、Cleaning of infant bed

Lift the plexiglass incubator hood, use the support locking plate to tilt the hood, then lift the infant bed and pull it out from the front along the track.

Wash all infant bed surfaces completely with the sanitizer, and then dry with the clean cloth; the mattress is removed, and covers again after changing, cleaning and drying.

2、Cleaning of bed layer board and air circulation

Take out the bed support and clean all its surfaces with detergent. Then use detergent to clean all the surfaces in the water trough bracket, heating tube, air temperature sensor (to wipe its surface lightly), axial fan casing and fan blades, air inlet pipe and air circulation channel, and then use a clean cloth wipe dry. As shown in the figure below, pull up the fan to remove the fan, soak it with a soft cloth with disinfectant, wring it out fully, and wipe all external surfaces.



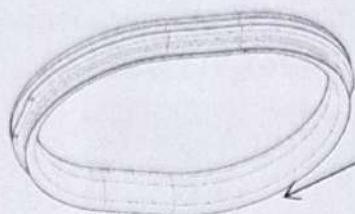
Be careful not to soak the fan with disinfectant, so as not to damage it



The temperature of the heater is high, and the heater can be cleaned at least 45 minutes after shutting down to avoid burns.

3、Cleaning of sealing rubber ring

Remove the sealing rubber ring from the plexiglass thermostatic cover, wash it with detergent and dry it.



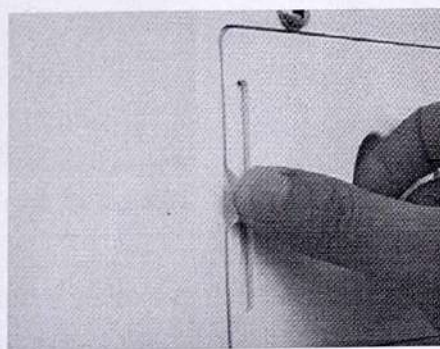
When installing, pay attention that this side faces outside the hood

4. Cleaning and replacement of air filter cloth

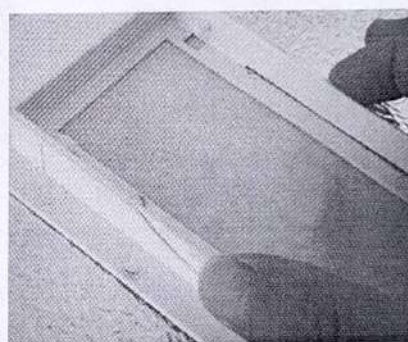


The filter cloth must use the special filter cloth equipped by our company, otherwise the air filtering effect will not be achieved or the carbon dioxide concentration of the incubator may exceed the standard.

Air filter cloth is generally replaced every three months. If it is found to be discolored or washed frequently, wrinkled and damaged, the filter cloth should be replaced in time. Press and hold the lock switch on the cover of the air filter box, and push to the right to open the air purifier box cover behind the incubator box. Squeeze the air filter cloth fixing platen from the outside inward, and take out the filter cloth. Clean the purifier box with a cleaning disinfectant, and then wipe it dry with a clean cloth. Clean the air filter cloth. If the filter cloth is cleaned frequently, wrinkles and breakage, the filter cloth should be replaced in time. First buckle one side of the air purifier box cover into the air purifier box, and press the other side firmly into the air purifier box to complete the cleaning.



(a)



(b)

Fig9-1 Open the air filter box operation

5. Cleaning of the skin temp sensor

Thoroughly clean the surface with a disinfectant cleaner and wipe dry with a clean cloth.

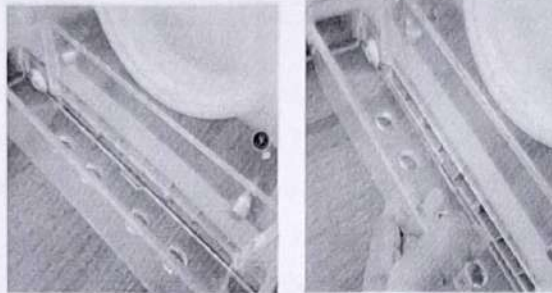
6. Cleaning of water trough

Pull out water trough, pour the water remained, wash water trough and each surface of the water trough support with the sanitizer and dry it, filling water trough with water, when finishing, pull off the water, load the water trough into support finally.

7. Cleaning of hood and the foot wheel(castors)

Wash all surfaces completely with the sanitizer, including corner and border on the door, and then wipe with the soft cloth lightly. Don't scrub the hood with organic solvents, such as alcohol, etc. Don't let the hood in ultraviolet direct radiation.

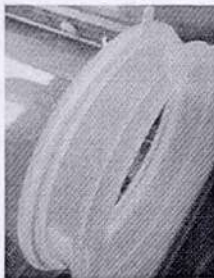
Cleaning of air outlet and air inlet guardrail



Remove the guardrail

Lift the guardrail slightly upward so that the guardrail fixing screw corresponds to the large end of the guardrail gourd hole, and then the guardrail can be removed. After cleaning, the corresponding hole corresponds to the fixing screw on the hood and then gently moves down to complete the installation.

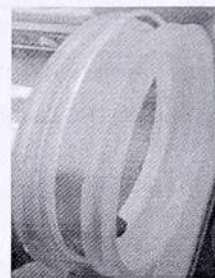
Cleaning of iris ports:



Remove the moving



Remove the rubber band of iris



Remove the fixed

Remove the iris ports sleeve and the moving coil, and then remove the rubber band of iris ports, the fixed coil can be canceled. After cleaning, reset the installation. Note: The rubber band for the iris ports must be installed in place. Forcefully select the fixed coil, and the fixed coil is not fixed. No falling off means that the rubber band is installed in place, and the rubber band cannot be partially raised after Installation



Install rubber bands for iris ports



Install the moving coil and sleeve

installation diagram of iris ports

9.2 Maintenance



Need professional personnel to maintain. While maintain, must pull out the power plug

1、The replacement of the battery

The service life of rechargeable batteries is generally 3 years. When performing power failure alarm inspection, keep the equipment running for 12 to 24 hours and unplug the power cord. If the duration of the power interruption alarm cannot reach 10 minutes or the sound pressure level of the alarm outside the incubator is less than 65dB, the battery should be replaced in time. The replacement of the battery must be carried out by authorized and qualified service personnel. If the battery fluid is found to flow out, it must be cleaned with detergent to avoid corrosion of the equipment. When replacing the battery, first unplug the power cord, then lift the plexiglass hood, use the support locking plate to tilt the hood, then lift the infant bed and pull it out from the front along the track. Take out the bed support, and then press down on one side of the shielding plate to tilt it, and then take out the shielding plate; The rechargeable battery is located on the largest circuit board (see the picture below). Solder it down, and solder a new battery with the model specification of B80H7A2H 8.4V to the circuit board according to the marked polarity. Don't throw away the replaced batteries randomly. Collect them together to avoid environmental pollution.



If the equipment hasn't been used over six months, start the machine to make it charge automatically at least 6 hours, insuring the sufficiency of power to alarm for power failure.

2、The replacement of the power fuse

Unplug the plug in the input socket on the power socket board on the right side of the control box, pull out the lower fuse holder, use a small screwdriver to push out the two fuse tubes one by one, find out the burned fuse, and replace it with the same After the new fuse tube of model specification (F6.3AL 250V, $\phi 5 \times 20\text{mm}$), push the fuse tube holder into place.

Replacement of transformer fuse tube. The fuse tube is located under the control box. Unscrew the fuse holder cover, take out the broken fuse tube, and replace it with a new fuse tube of the same model specification (F1AL 250V, $\phi 5 \times 20\text{mm}$).

Under normal circumstances, the incubator needs to be maintained every three months.

10 Common troubleshooting

Chart10-1 Baby incubator troubleshooting table

Trouble phenomenon	Trouble phenomenon	Handling method
Power failure alarm	The power socket is not well connected The power fuse is blown Grid power failure	Repair or replace the power socket Replace the fuse tube Turn off the power switch
Over-temp alarm	1、Temperature sensor is damaged	Replace temperature sensor
Temperature deviation alarm	Start the machine temperature on the low side The front door of the thermostatic cover has been opened for too long Set to a lower temperature when the air temperature is higher	Reset, wait Close the door and wait for reset Reset, wait
Sensor failure alarm	The sensor plug falls off or has poor contact 2.The sensor is broken, short-circuited or damaged	1.Install the sensor 2.Connect the broken wire or replace the sensor
Set key operation out of control	Key connect not good or damaged	Check the key button and deal with it
Humidity temp can't rise	water in tank too few humidity heat tube failure humidity already balance	add enough distilled water in tank check the heat tube, remove the failure adjust humidity rotary knob
The heater indicator is on, but temp has not been rising all the time	The heater is damaged Relay connect not well	Replace the heater Replace the relay

11 Circuit schematic

DC regulated power supply: 5V voltage for single-chip microcomputer and integrated circuit, 9V voltage for relay and buzzer.

8.4V built-in battery power: alarm power when the power is interrupted.

Heating control: It is composed of photoelectric coupling integrated block, thyristor, and 400W electric heating tube.

Alarm: constituted by a buzzer

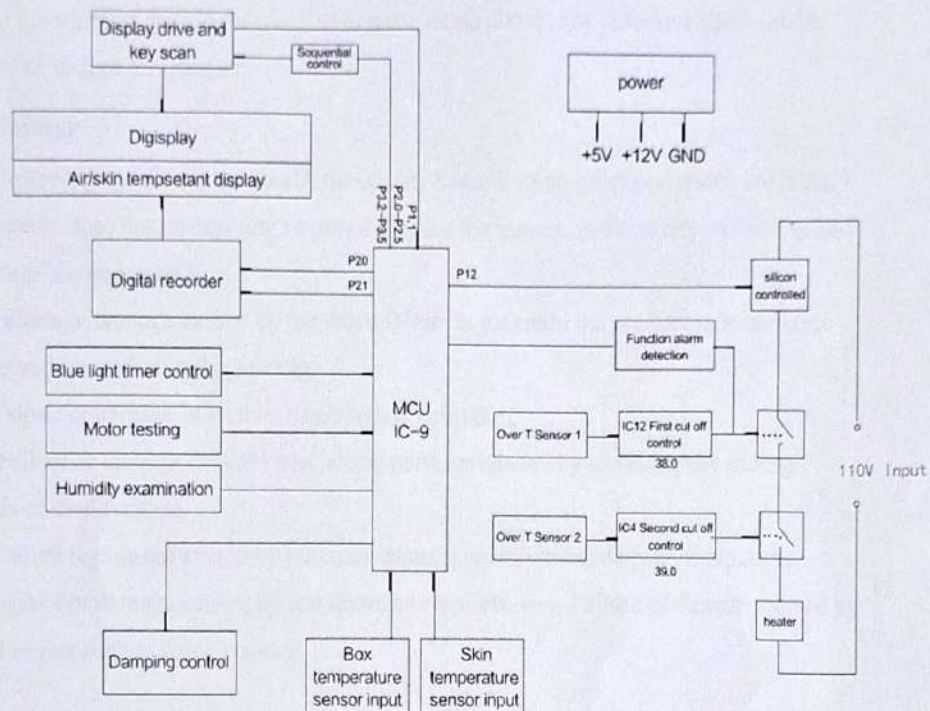


Figure 11-1 Hardware schematic diagram

If necessary, you can provide circuit diagrams, component lists, legends, calibration rules, or other information necessary to help the user's qualified technicians repair the repairable equipment parts designated by the manufacturer.

12 Quality Commitment and Disclaimer

12.1 Quality promise

If the products described in this manual are defective in materials and workmanship, they are guaranteed for one year from the date of leaving the factory, except for the following conditions:

1. All consumables and disposable items are only guaranteed free of charge for defects in delivery.

2. Confirm that it is normal maintenance and not included in the 1-year warranty period.

During the warranty period, in addition to those listed above, any defective parts can be replaced free of charge for the user.

12.2 Disclaimer

If the following conditions are found, the quality commitments proposed above are invalid, and the company does not assume any responsibility for the impact on the safety, reliability and performance of the equipment:

—— Failure or damage caused by the user's failure to maintain the product in accordance with the methods specified in this manual.

—— Failure or damage caused by user's wrong operation.

—— Failure or damage caused by not using parts designated by our company during maintenance or modification.

—— Failure or damage caused by purchase through unauthorized dealers or repairers.

—— Failure or damage caused by unauthorized repairers.—— Failure or damage caused by unexpected events such as force majeure

13 Packing list

Please check the accessories and spare parts listed in the list when unpacking.

Chart 13-1 Packing list

No	Name	Unit	Qty	Remarks
1	Control cabinet and cabinet	Pieces	1	
2	Hood	Set	1	
3	Infusion support with tray module	Set	1	Includes: infusion hook, infusion rod, infusion support and tray
4	Input power cord	Set	1	
5	Casters	Set	4	Including: 2 brake casters
6	Fuse (F6.3AL, $\phi 5 \times 20\text{mm}$)	Set	4	Spare
7	Fuses (F1AL, $\phi 5 \times 20\text{mm}$)	Set	2	Spare
8	Air purification filter cloth (dedicated)	Set	2	Spare
9	Certificate	Pieces	1	
10	User manual (including packing list)	Pieces	1	

14 After-sales service

Dear user:

Thank you for using our company's medical device products. If the product does not meet the technical indicators or other quality problems, please contact the company's after-sales service department to solve the problem according to the regulations.

15 Electromagnetic Compatibility

1、For this equipment, special precautions regarding electromagnetic compatibility (EMC) must be taken, and it must be installed and used in accordance with the electromagnetic compatibility information specified in this manual.

Portable and mobile radio frequency communication equipment may affect this equipment.

The cables and accessories provided by this device must be used. The cable information is as follows:

Chart 15-1 Cable information

Cable name	length
Power cord (10A)	2 meter

3、Except for cables (transducers) sold as spare parts for internal components, the use of accessories and cables (transducers) other than those specified may result in an increase in the emission of the equipment or system or a decrease in immunity.

4、The equipment or system should not be used close to or stacked with other equipment. If it must be used close or stacked, it should be observed to verify that it can operate normally under the configuration used.

5、Basic performance

Chart 15-2 Basic performance

Name	Specific description	Testing and verification methods
Infant incubator temperature display accuracy	In a stable temperature state, the difference between the average display temperature and the average incubator temperature should not exceed 0.8°C . (Article 50.106 of the special standard IEC60601-2-19)	In a stable state (set temperature: 36°C), the standard thermometer is placed at point A. During or after the experiment: The difference between the average incubator temperature and the average incubator temperature shall not exceed 0.8°C .

Schedule 1 Guidance and Declaration-Electromagnetic Emissions

Guidelines and manufacturer's declaration-electromagnetic emissions		
The infant incubator is expected to be used in the electromagnetic environment specified below, and the purchaser or user should ensure that it is used in such an electromagnetic environment.		
Launch experiment	Compliance	Electromagnetic environment-guide
GB 4824 (CISPR 11) Radio frequency emission	team 1	The baby incubator uses radio frequency energy only for its internal functions. Therefore, its radio frequency emission is very low, and the possibility of causing interference to nearby electronic equipment is very small.
GB 4824 (CISPR 11) Radio frequency emission	Class A	Infant incubators are suitable for use in non-domestic and all facilities that are not directly connected to the public low-voltage power supply network of households.
Class A (CISPR 11)	not applicable	
GB 17625. (IEC61000-3-3) Voltage fluctuation / flicker emission	not applicable	

Schedule 2 Guidance and Declaration-Electromagnetic Immunity

Guidelines and manufacturer's declaration-electromagnetic immunity			
The infant incubator is expected to be used in the electromagnetic environment specified below, and the purchaser or user should ensure that it is used in this electromagnetic environment.			
Immunity test	IEC60601 Test level	Coincidence level	Electromagnetic environment-guide
Electrostatic discharge (ESD) GB/T 17626.2 17626.2 (IEC61000-4-2)	±6kV contact discharge ±8kV air discharge	±6kV contact discharge ±8kV air discharge	The floor should be wood, concrete or ceramic tiles. If the floor is covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient burst GB/T 17626.4 (IEC61000-4-4)	±2kV to power line ±1kV to input/output line	±2kV to power line ±1kV to input/output line	The network power supply should have the quality used in a typical commercial or hospital environment.
Surge GB/T 17626.5 (IEC61000-4-5)	±1kV differential mode voltage ±2kV common mode voltage	±1kV differential mode voltage ±2kV common mode voltage	The network power supply should have the quality used in a typical commercial or hospital environment.
Voltage sag, short interruption and voltage change on the power input line GB/T 17626.11 (IEC61000-4-11)	<5% UT to 0.5 week (at UT, >95% dip) 40% UT vs. 5 weeks (at UT, 60% dip) 70% UT vs. 25 weeks (at UT, 30% dip) <5%UT to 5 s (at UT, >95% dip)	<5% UT to 0.5 week (at UT, >95% dip) 40% UT vs. 5 weeks (at UT, 60% dip) 70% UT vs. 25 weeks (at UT, 30% dip) <5%UT to 5 s (at UT, >95% dip)	The network power supply should be of the quality used in a typical commercial or hospital environment. If the user of the infant incubator needs continuous operation during the power interruption, it is recommended that the infant incubator use uninterrupted power supply or battery power supply.
Power frequency magnetic field (50Hz/60Hz) GB/T 17626.8 (IEC 61000-4-8)	3 A/m	3 A/m	The power frequency magnetic field should have the level characteristics of the power frequency magnetic field in a typical place in a typical commercial or hospital environment.
Note: UT is the AC grid voltage for which the test level is preferentially applied.			

Schedule 3 Guidance and Declaration-Electromagnetic Immunity

Guidelines and manufacturer's declaration-electromagnetic immunity			
The infant incubator is expected to be used in the electromagnetic environment specified below, and the purchaser or user should ensure that it is used in this electromagnetic environment.			
Immunity test	IEC60601 Test level	Coincidence level	Electromagnetic environment-guide
Conducted radio frequency GB/T 17626.6 (IEC61000-4-6)	3V (Effective value) 150kHz-80MHz	3V (Effective value)	Portable and mobile radio frequency communication equipment should not be used near any part of the incubator, including cables, than the recommended isolation distance. The calculation of this distance should use a formula corresponding to the transmitter frequency. Recommended isolation distance $d=1.2\sqrt{P}$ $d=1.2\sqrt{P}$ 80MHz~800MHz $d=2.3\sqrt{P}$ 800MHz~2.5GHz P is the maximum output rated power of the transmitter provided by the transmitter manufacturer, in watts (W), and d is the recommended isolation distance, in meters (m). The field strength of the fixed radio frequency transmitter is determined by the electromagnetic field survey a, and in each frequency range b should Interference may occur in the vicinity of equipment marked with the following symbols.
Radiated radio frequencyGB/T 17626.3 (IEC61000-4-3)	3V/m 80MHz-2.5GHz	3V/m	

Note 1: At 80 MHz and 800 MHz, the formula for the higher frequency band should be used.

Note 2: These guidelines may not be suitable for all situations. Electromagnetic propagation is affected by absorption and reflection from buildings, objects and humans.

a The field strength of fixed transmitters, such as base stations of wireless (cellular/cordless) telephones and ground mobile radios, amateur radio, AM and FM radio broadcasting and TV broadcasting, cannot be accurately predicted theoretically. In order to assess the electromagnetic environment of fixed radio frequency transmitters, electromagnetic field surveys should be considered. If the measured field strength of the infant incubator is higher than the above RF compliance level, the equipment should be observed to verify its normal operation. If abnormal performance is observed, supplementary measures may be necessary, such as reorienting or positioning the device.

b In the entire frequency range of 150KHz~80MHz, the field strength should be lower than 3 V/m.

Schedule 4 Recommended isolation distance between portable and mobile radio frequency communication equipment and equipment

Recommended isolation distance between portable and mobile RF communication equipment and equipment			
The infant incubator is expected to be used in an electromagnetic environment where radiated RF disturbances are controlled. According to the maximum output power of the communication device, the purchaser or user can prevent electromagnetic interference by maintaining a minimum distance between the portable and mobile radio frequency communication device (transmitter) and the infant incubator.			
Transmitter rated maximum Output power / W	Corresponding to the isolation distance of the transmitter at different frequencies/m		
	150kHz-80MHz $d=1.2\sqrt{P}$	80MHz-800MHz $d=1.2\sqrt{P}$	800MHz~2.5GHz $d=2.3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
For the transmitter rated maximum output power not listed in the above table, the recommended isolation distance d, in meters (m), can be determined by the formula in the corresponding transmitter frequency column, where P is the transmitter provided by the transmitter manufacturer The maximum output rated power, in watts (W).			
Note 1: At 80MHz and 800MHz frequencies, the formula for the higher frequency band should be used.			
Note 2: These guidelines may not be suitable for all situations. Electromagnetic propagation is affected by absorption and reflection from buildings, objects and human bodies.			